

CAO MANH TAN

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EDUCATION

Ha Noi University of Science & Technology (HUST)

Ha Noi, Viet Nam

Bachelor of Computer Engineering

Sep 2022 – Present

Relevant Coursework: System Analysis and Design, Object-Oriented Programming (OOP), Data Structures and Algorithms, Database Systems, Web Technologies and Cloud Services.

WORK EXPERIENCE

TECHTACK LIMITED COMPANY

Ha Noi, Viet Nam

Software Engineer Intern

Feb 2026 – May 2026

- **ENSOGO – Internal Human Resources Management Web & App:**
 - **Frontend Development (ReactJS, React Native, TypeScript, Ant Design):** Built responsive UI and complex multi-criteria data tables with server-side pagination, sorting, and debounce search.
 - **Backend & Database (Node.js, MySQL):** Implemented RESTful APIs and optimized relational queries with composite indexing, cutting query response latency by **40%** for large employee datasets.
 - **Form Management:** Designed reusable modal forms with strict validation workflows for leave management and staff onboarding.
- **TechTack Internal – Time-Tracking & Attendance App:**
 - **Mobile Attendance (React Native, Expo):** Designed cross-platform weekly timesheet and attendance tracking views with real-time total hours calculation.
 - **API Layer & CI/CD (Axios, JWT, Git Flow):** Standardized Axios interceptors for centralized error handling and automated token management; resolved 20+ peer code reviews.

FEATURED PROJECTS

Travel Stay & Hospitality Reservation Platform (Graduation Project)

Jun 2026 – Present

- **Full-Stack Architecture:** Designed a scalable backend following **Domain-Driven Design (DDD)** on **NestJS 11** & **TypeScript** (Controllers, Services, Repositories) paired with a **Feature-Based** modular frontend (app, modules, shared) on **ReactJS (Vite)**; enforced strict **DTO** validation via **class-validator** and documented endpoints using **Swagger / OpenAPI 3.0**.
- **Concurrency Control:** Implemented **Atomic Database Transactions** using **typeorm-transactional** on **MySQL 8.0**, completely eliminating race conditions and preventing double-booking during concurrent checkouts.
- **Real-Time Payments:** Automated dynamic VietQR payments with **payOS**, verified asynchronous **SHA-256 Webhooks**, and streamed real-time booking confirmation to the UI via **Server-Sent Events (SSE / EventSource)**, eliminating polling overhead.
- **Client State & Performance:** Optimized caching and query invalidation using **TanStack Query v5** (slashing redundant requests by **60%**); managed global multi-step booking drafts (date range, guest counters, price breakdown) across route changes using **Zustand**.
- **Auth & Navigation Security:** Built a dual **JWT** flow (**Access & Refresh tokens**) with a **Token Blacklist** table in **MySQL** for immediate logout revocation and **RBAC**; secured routes using **Axios Interceptors** and reusable **<ProtectedRoute />** components.
- **Geospatial UX:** Integrated interactive **Leaflet (react-leaflet)** maps with **LocationIQ / Goong Maps** geocoding APIs for address search, coordinate pinpointing, and visual homestay discovery.

- **Interactive Ticketing UX:** Developed dynamic 2D seat layout matrices (single/double-deck) with real-time seat locks and dynamic price calculation using **React 19** and **Ant Design**.
- **Real-Time Chat & High-Performance Data Layer:** Built bidirectional support chat via **Socket.IO** and **NestJS Gateway**; optimized relational queries with custom **TypeORM QueryBuilder** and **MySQL** to eliminate eager-loading overhead.
- **Authentication & Payments:** Integrated 6-digit email OTP recovery via **Resend API**, dynamic QR checkout via **payOS VietQR**, and full API documentation with **Swagger**.

SKILLS & CERTIFICATIONS

Frontend: ReactJS (18/19), React Native, TypeScript, JavaScript (ES6+), HTML5, CSS3, Ant Design, Leaflet Maps, React Router v7.

Backend & Real-time: Node.js, NestJS 11, Express, RESTful APIs, Server-Sent Events (SSE), WebSocket / Socket.IO, Domain-Driven Design (DDD).

Database & ORM: MySQL 8.0, MongoDB, TypeORM (Custom QueryBuilder, Atomic Transactions).

State & Tools: Zustand, TanStack Query v5, Axios, Git, Postman, Swagger / OpenAPI 3.0, Jira.

Security & Integrations: Dual JWT, Token Blacklist, RBAC, payOS (VietQR), Resend, Nodemailer, Cloudinary.

Languages: Vietnamese (Native), English (Reading Technical Documentation, Basic Communication).

Certifications: TOEIC 755/990 (Listening & Reading).